EEET4000 - Engineering Research Project 1
Milestone Report

By Lucas Lim (20841085)

Table of Contents

Table of Figures III
List of Tables III
1.0 Introduction and Background 1
1.1 Abstract 1
1.2 Introduction 1
2.0 Literature Review Summary 3
2.1 What and Why do Hallucinations Occur 3
2.2 Types of Hallucinations 3
2.3 Model Designs that Counter Hallucinations 4
2.4 Evaluation Metrics for Hallucination Detection and Model Reliability 5
3.0 Milestone/Progress Achieved 7
3.1 Methodology 7
3.2 Baseline RAG Model 10
3.3 Test Set Questions 13
3.4 Evaluation Metrics 15
4.0 Sustainability Considerations 17
4.1 Environmental Sustainability 17
4.2 Economic Feasibility 18
4.3 Technical Sustainability 18
4.4 Social Responsibility & Ethical Considerations 19
4.5 Overall Sustainability Impact 19
5.0 Conclusions and Project 2 Plan 20
6.0 References 21

Table of Figures

Figure 1: Snapshot of code for extracting text from PDF governance documents 10
Figure 2: Snapshot of code for creating chunks from the extracted text 10
Figure 3: Snapshot of code for creating a vector database 11
Figure 4: Snapshot of code for creating embeddings from the chunks 11
Figure 5: Snapshot of code for passing prompt and context to Ollama LLM 11
Figure 6: Snapshot of code for retrieving model response 12
Figure 7: Snapshot of example output from test question 12

List of Tables

Table 1: List of model configurations for future experimentation 8
Table 2: Summary, description, and examples of questions in test set for future evaluation and experimentation 13

1.0 Introduction and Background

1.1 Abstract

This project investigates methods for reducing the risk of hallucinations in Large Language Model (LLM) based question answering (QA) systems operating on governance documentation. Following the completion of the literature review, a baseline Retrieval-Augmented Generation (RAG) model has been developed using governance-related IEEE documents as the primary governance corpus. The current progress includes document pre-processing, vector database implementation, retrieval pipeline development, and the creation of a benchmark question set for future evaluation. This project also identifies and prepares multiple hallucination mitigation strategies for future experimentation, including Reinforced Learning with Self Verification (RISE), Reinforced Learning with Verifiable Rewards (RLVR), Explainable Artificial Intelligence (XAI), and multi-modal verification approaches. Evaluation methodologies involving hallucination rate, faithfulness, retrieval quality, and traditional performance metrics have also been established. Future work will focus on integrating and comparing these mitigation strategies to determine their effectiveness in improving factual accuracy, retrieval reliability, and overall trustworthiness within governance-oriented QA systems.

1.2 Introduction

Large Language Models (LLMs) have become increasingly prominent, especially in modern question answering (QA) systems due to their ability to generate natural, contextually relevant, and human-like responses. Popular models such as ChatGPT by OpenAI, Gemini by Google, Grok by xAI, and Claude by Anthropic are now widely used across a range of professional, educational, and personal applications. Despite their capabilities, concerns regarding their ability to remain consistent, factually reliable, and trustworthy, particularly in domains where accuracy is absolutely critical. One major issue associated with LLMs is the occurrence of hallucinations, where responses generated by the model may contain false, misleading, or unsupported information [1].

Hallucinations in generative AI can arise from a variety of factors. These might include inaccuracies within the training datasets, limitations in the probabilistic text generation, and insufficient grounding in reliable source material. Since LLMs predict responses based on learned language patterns rather than verified factual reasoning, they may produce outputs that appear convincing despite lacking factual support. This presents a significant issue for applications concerning governance documentation, where the incorrect interpretations of constitutions, bylaws, or operational procedures may lead to misinformation or procedural inconsistencies [1].

Existing research has proposed several hallucination mitigation strategies used in LLM-based QA systems. These include Retrieval Augmented Generation (RAG), Reinforced Learning with Verifiable Rewards (RLVR), Reinforced Learning with Self Verification (RISE), Explainable AI (XAI), extractive approaches, and multi-modal verification frameworks [1]. Current literature suggests that no single technique is sufficient to fully eliminate hallucinations, and that combinations of retrieval, verification, and explainability strategies are often required to improve the reliability of responses.

Building upon this research, the current project focuses on the development and evaluation of governance-oriented QA systems designed to reduce hallucinations within generated responses. A baseline RAG model has been implemented using IEEE governance documentation, including constitutions, bylaws, and operational manuals, as the primary knowledge corpus. Current progress includes document pre-processing, vector embedding generation, vector database integration, retrieval pipeline implementation, and the development of the benchmark question set consisting of governance-related queries. Future work will involve integrating additional hallucination mitigation strategies, including RISE, RLVR, XAI, and multi-modal verification techniques, followed by a comparative analysis using retrieval quality, factual consistency, hallucination rate, and computational efficiency evaluation metrics.

2.0 Literature Review Summary

2.1 What and Why do Hallucinations Occur

Artificial Intelligence (AI) and Large Language Model (LLMs) has had many advancements over the past decade, enabling increasingly human-like responses and interactions across many industries and applications. However, the growing implementation of AI in critical areas such as healthcare, governance, law, and education has raised concerns regarding the reliability and accuracy of AI generated responses. One of the most significant and prominent issues associated with LLMs is hallucination.

Hallucination occurs when a model generates information in its responses that appears to be convincing and accurate but is actually false, misleading, fabricated, or unsupported by factual evidence. This exists because LLMs generate responses by predicting statistically likely word sequences rather than verifying the accuracy and truthfulness of the information. As a result, models may end up producing incorrect outputs that they are confident in, particularly when dealing with incomplete information, ambiguous prompts, or unfamiliar contexts.

Hallucinations are further influenced by factors such as incomplete training data, overgeneralisation, and limitations in the model's reasoning ability. Since LLMs tend to rely heavily on learned patterns within their training datasets, they may attempt to fill information gaps even if there is no available source of reliable knowledge. Consequently, hallucinations remain one of the major limitations of AI models that prevent full trust in generative AI systems.

2.2 Types of Hallucinations

Hallucinations can appear in a variety of forms depending on the type of error generated by the AI model. One very common issue is overfitting, where the model memorises the training data too specifically instead of learning the broader more general patterns in the date. Although this may lead to improved performance on the known training data, it often causes poor generalisation when handling new or unseen inputs.

Other types of hallucinations include logic and reasoning errors. Logic errors occur when there exists no proper causal relationship between the generated response with the input query, while reasoning errors arise when the model fails to understand human behaviour, humour, sarcasm, metaphors, or social contexts [2]. These limitations demonstrate that LLMs can imitate conversation without fully comprehending human experience and intent.

Mathematical hallucinations are also common, particularly in queries involving calculations, numerical reasoning, geometry, and unit conversions. In addition, models may also generate unfounded fabricated information such as fictional references, false medical advice, or invented events. Factual errors can also occur when the output directly contradicts the truth and commonly known knowledge [2].

Finally, text output errors include spelling mistakes, grammatical issues, syntax problems, and translation inaccuracies. While less severe and damaging than factual hallucinations, these errors still lead to the reduction of the credibility and usability of AI-generated

responses [2]. Overall, hallucinations highlight the limitations of LLMs in ensuring truthfulness, consistency, and contextual understanding despite their ability to generate fluent and realistic language.

2.3 Model Designs that Counter Hallucinations

Due to the increasing importance of reliable AI systems, several approaches have been developed to reduce hallucinations and improve the accuracy of responses. One of the most widely used techniques is Retrieval Augmented Generation (RAG). RAG systems improve reliability by retrieving relevant documents or external information before generating a response to a query [3]. Instead of relying solely on the model's internal knowledge, the retrieved data provides grounding for the generated output. This in turn will significantly reduce the likelihood of fabricated information and improves factual accuracy [4]. However, RAG systems depend heavily on the quality of the retrieved documents and may still struggle with poorly structured queries or privacy concerns involving sensitive data.

Another approach is Reinforcement Learning with Self Verification (RISE), where the model internally evaluates its own generated responses to improve its accuracy over time [5]. Similarly, Reinforcement Learning with Verifiable Rewards (RLVR) uses external verification systems to assess response correctness [6]. Both of these methods aim to reduce hallucinations by validating outputs. However, they also increase computational complexity and depend heavily on the effectiveness of their verification mechanisms.

Extractive AI offers a different solution by avoiding the entire generative process altogether. Instead of creating new responses, extractive systems directly retrieve and present information from source documents [7]. This greatly reduces hallucinations as there can be no additional information being generated. However, extractive AI lacks flexibility and conversational capability, making it much less effective for complex or ambiguous queries and human-like QA. Another developing technique is multimodal verification, which validates information using multiple sources [8]. These might include text, images, audio, or video. Cross-checking information across modalities can improve robustness and reduce inconsistencies, with the downside of being more computationally expensive and technically challenging.

Finally, Explainable AI (XAI) focuses on improving transparency rather that working to directly prevent hallucinations. XAI systems attempt to explain to the user how decisions and responses are generated, allowing users to better understand the reasoning process and identify potential inconsistencies or unsupported claims [9]. This improves trust and accountability, particularly in high-risk applications.

Overall, current research demonstrates that no single solution completely eliminates hallucinations without trade-offs. Techniques such as RAG, RISE, RLVR, extractive AI, multi-modal verification, and XAI each improve reliability in different ways while introducing their own limitations in flexibility, complexity, scalability, or interpretability. As a result, hallucination reduction remains to be an active and increasingly important area of AI research.

2.4 Evaluation Metrics for Hallucination Detection and Model Reliability

Evaluating the reliability and correctness of AI-generated responses is essential in determining the overall effectiveness of a model in QA systems. Common evaluation metrics include accuracy, precision, recall, and F1-score, each measuring different aspects of model performance [10]. Accuracy measures the proportion of correct predictions made by the model. While a high accuracy score may be an indication of a strong overall performance, it is often unreliable in real-world applications because it does not account for imbalanced datasets, noise in data, or changing environments. As a result, accuracy alone can end up overestimate a model's true effectiveness.

Precision and recall provide more detailed evaluations. Precision is a measure of how many predicted positive results are actually correct, making it useful in situations where false positives are costly, such as fraud detection or facial recognition. Recall measures how effectively the model identifies all actual positives instances and becomes important in applications such as medical diagnosis or security systems [10]. These metrics often exist in a trade-off relationship, where improving one can reduce the other.

The F1-score combines precision and recall into a single balanced metric, which makes it particularly useful for the evaluation of systems with imbalanced data [10]. A high F1-score indicates that both precision and recall are performing well. However, although these traditional metrics provide useful insights, they still fail to fully comprehend contextual relevance robustness, and the practical impact of hallucinations in generative AI systems.

To better evaluate hallucinations, researchers have developed faithfulness-based metrics that measure how grounded a generated response is in its source material. These metrics perform an assessment as to whether the output is supported by evidence rather than fabricated information. Common faithfulness metrics include Decision Flip -Most Informative Token (DFMIT), Decision Flip - Fraction of Tokens (DFFOT), Comprehensiveness (COMP), Sufficiency (SUFF), Correlation between Importance and Output Probability (CORR), and Monotonicity (MONO) [11]. These methods generally work by identifying important tokens within the input and measuring how changes to those tokens affect the model's predictions.

Although faithfulness metrics provide deeper insight into model reasoning and hallucination reduction, they also introduce their own set of challenges. Many rely on masking or removing tokens, which can create a new issue called Out-of-Distribution (OOD) inputs that do not resemble realistic data. This can lead to the formation of unstable or misleading evaluation results.

To address these issues, Fine-tuned Fidelity (F-Fidelity) was developed as an improved faithfulness evaluation metric. F-Fidelity reduces the OOD problem by applying controlled random masking and explanation-agnostic fine-tuning during its training and evaluation phases [12]. This improves the reliability and consistency of faithfulness assessments while reducing information leakage and bias.

Overall, traditional metrics such as accuracy, precision, recall, and F1-score remain useful for evaluating general predictive performance, while faithfulness-based metrics provide more

specialised insight into hallucination detection and interpretability. However, no single metric fully captures all aspects of real-world AI reliability, making evaluation an ongoing challenge in generative AI research.

3.0 Milestone/Progress Achieved

3.1 Methodology

After conducting research into the existing literature and research made on the topics related to my thesis, changes were made to my methodology and how the reduction of hallucinations are to be tested. This research will adopt an experimental comparative methodology to investigate the numerous techniques available for reducing hallucinations in LLM-based QA systems operating on governance documents. The study will focus on improving factual reliability and truthfulness in the responses generated from existing organisational documents by integrating hallucination mitigation strategies and technologies into a RAG framework. A standard RAG architecture is selected to be the baseline system due to its widespread adoption and demonstrated effectiveness in grounding generated responses using externally retrieved contextual information and data. Since the effectiveness of RAG systems has already been extensively researched within existing literature, this study will not seek to validate the RAG architecture itself and instead focusing on using the baseline RAG model as a controlled configuration against which additional hallucination reduction techniques are applied and evaluated.

This research will investigate whether supplementary strategies integrated into the RAG pipeline model can further improve factual consistency, retrieval quality, and overall reliability in its response when handling governance-related queries. The selected techniques include Reinforced Learning with Self Verification (RISE), Reinforced Learning with Verifiable Rewards (RLVR), Explainable AI (XAI), and Multi-modal verification mechanisms. The document corpus used throughout the experimentation process of the thesis consists of governance and operational documents sourced from IEEE and associated organisations, including the IEEE Constitution and Bylaws, Draft Operational Manual V5, SMC Constitution and Bylaws (2023), and the SMCS Technical manual (2023). These documents were selected due to their highly structure and policy-orientated nature, where the factual accuracy of information retrieved is essential. Governance documentation presents an appropriate environment for hallucination analysis because incorrect responses may lead to procedural misinterpretations, violations of the policy, or misinformation regarding any regulations of the organisation.

Prior to any experimentation, all documents are to undergo various pre-processing steps to prepare them for retrieval and generation tasks. This pre-processing stage includes text-extraction, normalisation, segmentation into smaller contextual chunks, and vector embedding generation. The processed embeddings are stored withing a vector database to enable semantic similarity retrieval during query execution. The baseline RAG architecture follows the conventional retrieval-generation pipeline. User queries are first converted into vector embeddings and compared against the vector database to retrieve the most relevant document chunks. The retrieved contextual information is then appended to the prompt supplied to the LLM, enabling a grounded response to be generated based on the governance corpus. This baseline configuration remains fixed throughout the experimentation process to ensure that consistency and fairness is achieved across all model comparisons.

Several enhanced model configurations are then developed by integrating additional hallucination mitigation strategies into the baseline architecture. Each experimental model

modifies the baseline RAG pipeline while maintaining the same dataset, question sets, prompting structure, and retrieval environment. This allows performance variations to be attributed specifically to the applied mitigation strategy rather than external experimental inconsistencies.

The Evaluated Configurations are summarised below:

Table 1: List of model configurations for future experimentation

Model Configuration	Description
Baseline RAG	Standard RAG Pipeline
RAG + RISE	Incorporates internal self-evaluation for iterative verification
RAG + RLVR	Applies external verifiable reward mechanisms
RAG + XAI	Integrates explainability techniques to improve reasoning transparency
RAG + Multi-modal Verification	Uses additional verification mechanisms across multiple contextual representations
RAG + Combined Techniques	Hybrid implementation combining multiple mitigation strategies

To evaluate performance, a benchmark question set consisting of approximately 50 to 100 governance-related queries is manually developed from the source documents. The questions are designed to represent multiple forms of governance inquiry. These might include factual retrieval questions, questions interpreting procedures, policy clarifications, and context-sensitive or multi-document reasoning tasks. This will ensure that the evaluation process measures both simple information retrieval and more complex reasoning capability. Each model configuration is tested using the identical question set under controlled experimental conditions. The responses generated are collected and compared against reference answers derived directly from the governance documents. To maintain consistency, prompting structure, retrieval parameters, chunk sizes, and generation settings shall remain standardised across all experiments made.

A formal definition of hallucinations is established to ensure consistency in the evaluation throughout the research. Within this context of study, a hallucination can be defined as any generated statement that is unsupported, contradicting, or unverifiable within any of the provided governance document corpus. Responses containing fabricated polices, incorrect procedural details, or unsupported interpretations are classified and labelled as hallucinated outputs by the model. Evaluation is conducted using both generation-based and retrieval-based evaluation and performance metrics. Generation-based evaluation metrics such as Accuracy and F1-score will be used to evaluate whether the generated answers are correct, verifiable, and properly grounded withing the governance documentation. Retrieval-based metrics such as top-k retrieval accuracy and Mean Reciprocal Rank (MRR), will be used to determine the effectiveness of the retrieval system in supplying relevant contextual information required for accurate answer generation.

In addition to correctness metrics, computational efficiency is also considered. Performance indicators such as response latency, memory consumption, and computational overhead are measured to evaluate the practical feasibilities of each of the mitigation strategies. This ensures that any improvements in factual reliability are assessed alongside operational efficiency and scalability of the model.

Following experimentation, comparative analysis is conducted across all model configurations. The analysis will focus on identifying any reductions in the frequency of hallucinations, improvements in factual correctness, retrieval effectiveness, interpretability, and computational performance of the model. Statistical comparisons and trend analysis are used to determine which enhancement strategy may provide the most effective balance between accuracy, truthfulness, retrieval quality, and system efficiency within governance-oriented QA systems.

There are certain issues to methodological rigor that should be noted. Firstly, the document corpus is limited to a relatively small collection of governance documentation, which may reduce the generalisability of findings to broader domains. Second, since the questions are developed manually, it may introduce a selection bias depending on how representative the questions are of real-world governance inquiry applications. Third, evaluation outcomes may be influenced by LLM-specific designs, as different language models may exhibit varying hallucination behaviours and reasoning capabilities. Lastly, engineering choices and retrieval parameter configurations may affect the behaviour of the model, potentially influencing the comparison between the different model's performance results.

To minimise these limitations, the project aims to maintain standardised experimental conditions across all model configurations and applies consistent evaluation procedures throughout the testing of the model's capabilities. Multiple quantitative metrics are used to reduce reliance on any one single performance indicator, while governance-derived reference answers are used to support objective assessment of factual correctness and hallucination occurrence.

3.2 Baseline RAG Model

A basic RAG model has been created to act as a baseline for all future experimentation. The pipeline of the model is as follows:

It starts by creating a list that stores the information extracted from the documents. The governance documents are all stored in a folder named docs in the same directory as the model. From this folder, the code loops through each PDF file and builds a path to each file. It then opens each of the files, creating an empty text variable, loops through each of the pages in the PDF and extracts the readable text from it. It then appends this to the documents list as the filename and the extracted text. This is shown below:

```
documents = []

folder = "docs"

for filename in os.listdir(folder):

    if filename.endswith(".pdf"):

        path = os.path.join(folder, filename)

        reader = PdfReader(path)

        text = ""

        for page in reader.pages:

            extracted = page.extract_text()

            if extracted:
                text += extracted

        documents.append({
            "filename": filename,
            "text": text
        })

print("Documents loaded:", len(documents))
```

Figure 1: Snapshot of code for extracting text from PDF governance documents

The next step is the chunking process of the extracted text. The chunk_text function works by creating a list, whilst looping through all the text, with the defined chunk sizes, it slices the text into the specified sizes and saves it into the list. These chunks are saved as the chunks and the documents the text that was chunked is from.

```
def chunk_text(text, chunk_size=800, overlap=200):

    chunks = []

    start = 0

    while start < len(text):

        end = start + chunk_size

        chunk = text[start:end]

        chunks.append(chunk)

        start += (chunk_size - overlap)

    return chunks


all_chunks = []

for doc in documents:

    chunks = chunk_text(doc["text"])

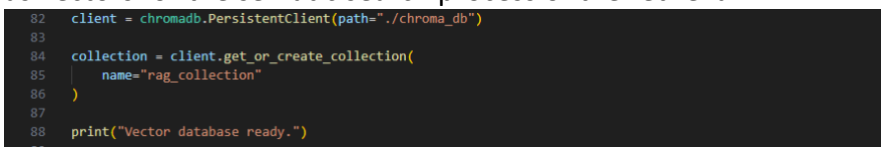
    for chunk in chunks:

        all_chunks.append({
            "text": chunk,
            "source": doc["filename"]
        })

print("Total chunks created:", len(all_chunks))
```

Figure 2: Snapshot of code for creating chunks from the extracted text

Next is creating a ChromaDB vector database in which the RAG system will store the chunks as vectors for the sematic search process of the retrieval.



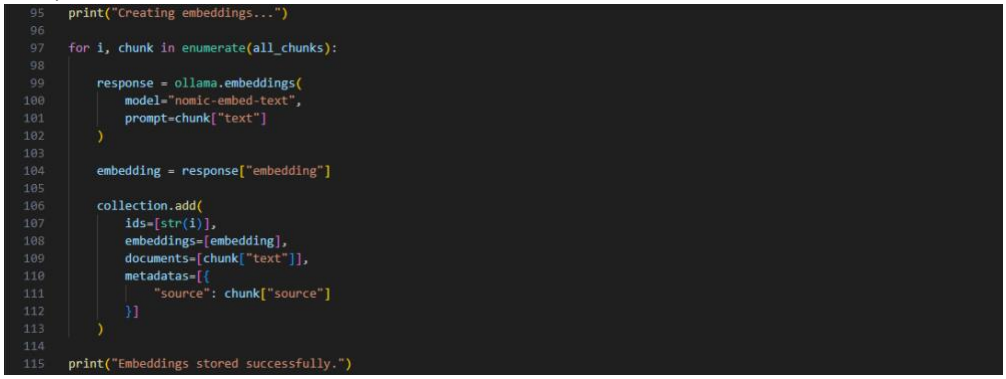
```
client = chromadb.PersistentClient(path="./chroma_db")

collection = client.get_or_create_collection(
    name="rag_collection"
)

print("Vector database ready.")
```

Figure 3: Snapshot of code for creating a vector database

After the database is made, loops through every chunk stored and adds an index to it. Using Ollama (an Open-Source AI tool that allows you to run LLMs locally), the chunk texts are sent into an embedding model. After retrieving the numerical embedding from the response object, the embedding and related information is then added into the vector database. In each addition into the database, it contains a unique ID, the embedding vector, the original text, and the metadata.



```
print("Creating embeddings...")

for i, chunk in enumerate(all_chunks):

    response = ollama.embeddings(
        model="nomic-embed-text",
        prompt=chunk["text"]
    )

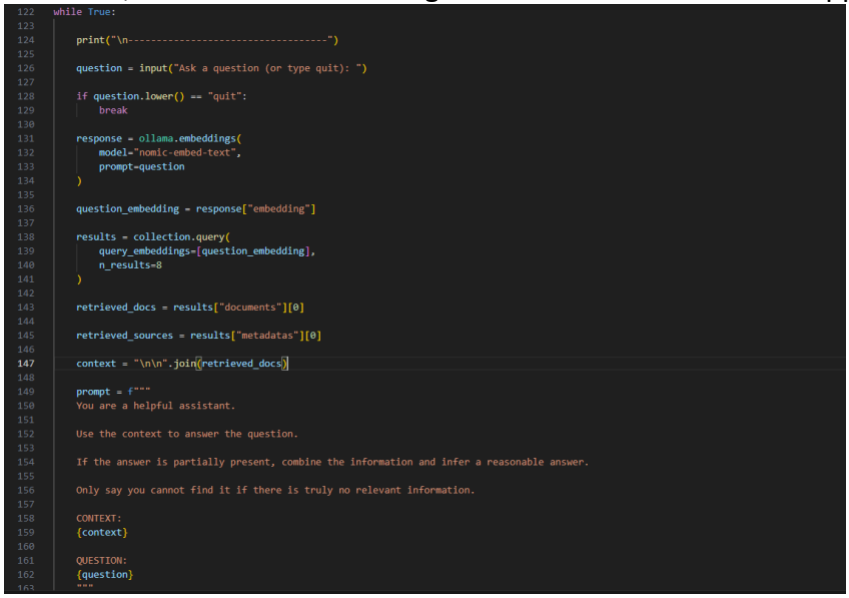
    embedding = response["embedding"]

    collection.add(
        ids=[str(i)],
        embeddings=[embedding],
        documents=[chunk["text"]],
        metadatas=[{
            "source": chunk["source"]
        }]
    )

print("Embeddings stored successfully.")
```

Figure 4: Snapshot of code for creating embeddings from the chunks

This section of the code is simply about the user's queries to the model. The question made by the user is converted into a vector and then the numerical vector representation of the question is stored. A semantic similarity search is then performed on the vector database in which the question embeddings are compared against all the stored chunk embeddings and then the most semantically relevant chunks are retrieved. In this current model, the top 8 most relevant chunks are retrieved. The actual text chunks and associated metadata are retrieved, combined into one large contextual block which is supplied to the LLM.



```
while True:

    print("\n----------------------------------")

    question = input("Ask a question (or type quit): ")

    if question.lower() == "quit":
        break

    response = ollama.embeddings(
        model="nomic-embed-text",
        prompt=question
    )

    question_embedding = response["embedding"]

    results = collection.query(
        query_embeddings=[question_embedding],
        n_results=8
    )

    retrieved_docs = results["documents"][0]

    retrieved_sources = results["metadatas"][0]

    context = "\n\n".join(retrieved_docs)

    prompt = f"""
    You are a helpful assistant.

    Use the context to answer the question.

    If the answer is partially present, combine the information and infer a reasonable answer.

    Only say you cannot find it if there is truly no relevant information.

    CONTEXT:
    {context}

    QUESTION:
    {question}
    """
```

Figure 5: Snapshot of code for passing prompt and context to Ollama LLM

From here, the Ollama chat API is called to generate a response from the language model. Specifying the model to be used for the response generation, the prompt, along with the contextual information is provided to the model. The generated text of the model's response is then retrieved and displayed in the terminal.

```python
response = ollama.chat(
    model="llama3",
    messages=[
        {
            "role": "user",
            "content": prompt
        }
    ]
)

answer = response["message"]["content"]

print("\nANSWER:\n")
print(answer)

print("\nSOURCES:\n")

for source in retrieved_sources:
    print("-", source["source"])
```

Figure 6: Snapshot of code for retrieving model response

Overall, the process goes from Governance PDFs -> Text Extraction -> Chunking -> Embedding Generation -> Vector Database Storage -> Question Embedding -> Semantic Retrieval -> Prompt Augmentation -> LLM Generation -> Answer + Source Output.

```
Ask a question (or type quit): What is the purpose of the IEEE Constitution and Bylaws?

ANSWER:

Based on the provided context from the IEEE Constitution and Bylaws, it appears that the purposes of these documents are:

* Scientific, literary, and educational in character (Article I, Section 2)
* To govern the operations and administration of the IEEE (Article II, Section 1)
* To maintain a high professional standing among members (Article I, Section 2)

In summary, the primary purpose of the IEEE Constitution and Bylaws is to establish the organization's goals, objectives, and governing structure, while also promoting scientific, literary, and educational activities.

SOURCES:

- ieee-constitution-and-bylaws.pdf
- SMC-Constitution-Bylaws-August-2023-web.pdf
- ieee-constitution-and-bylaws.pdf
- SMC-Constitution-Bylaws-August-2023-web.pdf
- ieee-constitution-and-bylaws.pdf
- draft Operational Manual V5.pdf
- SMC-Constitution-Bylaws-August-2023-web.pdf
- ieee-constitution-and-bylaws.pdf
```

Figure 7: Snapshot of example output from test question

This model is still an initial prototype. Ollama and ChromaDB are both used for their Open-source lightweight local deployment capabilities and compatibility with python-based vector retrieval pipelines. At this point in time, the ability to deploy the model locally seemed to be the most efficient rather than relying on cloud-based services. This is important as it not only saves in experimental costs but also allows for others to reproduce this experimentation without much difficulty. Most other factors such as chunk size and top-k retrieval were chosen after slight experimentation for best results, as well as the most commonly used variables.

Whilst possible to be used as the baseline RAG model, there are several factors that can still be fine-tuned. These might include the chunking of the texts from the governance documents, in which changing the size and overlay of the chunks might affect the overall quality of the retrieval process and contextual information. The number of chunks retrieved can also be altered. Increasing this number may lead to better context coverage and improved multi-document reasoning. However, having too many chunks may result in the introduction of irrelevant information as well as increased hallucination risk from noise along with the increased computational cost.

Finding an appropriate variable is important for balancing grounding, reasoning, and efficiency. Another change that could be made would be the prompt that is fed into the Ollama model to generate the response. The factors are still being experimentally evaluated to find the most optimal model design to act as the baseline before the addition and implementation of the other hallucination mitigation strategies.

3.3 Test Set Questions

For the testing set, there will be around 50-100 questions, all of which will be designed to represent various forms of governance inquiry. These questions aim to examine the model's ability to retrieve information from the governance documentation as well as testing the reasoning capabilities of the model. As such, the questions will include factual retrieval questions, interpretation and clarification of policies, and multi-document inquiries.

Questions can be placed into the following categories:

Table 2: Summary, description, and examples of questions in test set for future evaluation and experimentation

Category	Goal	Examples
Direct Retrieval Questions	Testing for whether the model is able to retrieve explicit information. These are necessary to test baseline retrieval accuracy.	What is the purpose of the IEEE Constitution and Bylaws? What is the role of the Operational Manual? Who is eligible for membership in the society? How are amendments approved?
Cross-Document Questions	Tests whether the RAG system can retrieve multiple relevant chunks and combine them without fabricating links. It is useful to expose weak retrieval processes and hallucinated combination information.	How does the Operational Manual support the rules defined in the Constitution and Bylaws? Which document defines committee responsibilities? What are the differences between the Society Constitution and the IEEE Constitution regarding governance?
Semantic Retrieval Questions	Testing the semantic understanding of the model instead of keyword matching, ensuring that the model understands	Who has authority over procedural decisions? What happens when a committee position becomes vacant?

	meaning rather than just the words.	What steps are required for advancement within the organisation?
Hallucination Trap Questions	Asking for information not found in the governance documentation to measure the model's resistance to hallucination.	What is the annual salary of IEEE officers? What colour scheme if officially required for committee branding? Does IEEE require AI ethics certification for members?
Ambiguous Question	Questions that check whether any improvements made reduce any overconfident hallucinations.	Who approves changes? How are committees managed?
Contradicting Questions	Testing that the reasoning of the model is grounded in the documents and that it is aware of its sources.	Which document takes precedence when procedures conflict? Are committee procedures identical across all documents?

3.4 Evaluation Metrics

For the evaluation phase of the experiment, the evaluation metrics serves to be used to measure effectiveness, reliability, and the overall practicality of the proposed hallucination mitigation strategies integrated into the RAG model framework. Since the objective of the study is not only to improve the overall accuracy of responses made by the model, but to also reduce any unsupported and fabricated information generated by the model, multiple categories of evaluation metrics are used to comprehensively evaluate the performance of the model's capabilities.

The selected metrics can be divided into three categories: response quality metrics, retrieval quality metrics, and computational efficiency metrics. Used together, these metrics aim to provide a balanced evaluation of how accurately the model is able to retrieve information from the governance corpus, how truthfully it is able to generate responses, and how efficiently the model operates under different hallucination mitigation strategies.

Response quality metrics are used to assess the factual correctness and reliability of the responses generated by the model. Metrics such as accuracy, precision, recall, and F1-score evaluate the model's ability to produce 'correct' responses whilst minimising any incorrect or misleading outputs. Accuracy measures the overall proportion of correctly generated responses. Precision evaluates how many generated responses are factually correct relative to the total number of generated responses. Recall is a measure of the system's ability to successfully identify and reproduce relevant information from the governance corpus. The F1-score provides a balanced evaluation between precision and recall.

In addition to conventional classification metrics, hallucination-specific metrics are incorporated due to the main objective of the study. Hallucination rate measures the frequency in which a response contains unsupported, fabricated, or unverifiable information relative to the documents found in the governance corpus. Faithfulness and factual consistency metrics are also used to determine whether generated outputs remain grounded in the retrieved contextual information supplied by the model. These metrics become especially important in governance-related QA systems, where procedural and policy related information that is incorrect may lead to serious interpretational issues.

Retrieval quality metrics are included to evaluate the effectiveness of the document retrieval process of the RAG model architecture. Since the responses generated by the model depend heavily on the quality of the information retrieved, metrics such as Recall@k, Mean Reciprocal Rank, retrieval precision, and context relevance are used to assess overall retrieval ability [13]. Recall@k measures whether the relevant supporting information appears within the top-k retrieved document chunks. MRR evaluates the process in which the highly relevant documents are ranked during the retrieval process [13]. Context relevant metrics further assess whether the retrieved passages can meaningfully contribute to accurate answer generation rather than introducing any irrelevant or misleading information.

Computational efficiency metrics are also incorporated to evaluate the practical efficiency of each of the hallucination mitigation strategies. While certain techniques may improve factual correctness, they may also inadvertently introduce increased computational

overhead or increased response latency. Therefore, metrics such as inference time, retrieval latency, token usage, memory consumption, and overall response generation time are measured throughout the experimentation. These metrics help determine whether improvement in hallucination reduction can be achieved in an efficient manner, enough so that it can be realistically deployed in real-world scenarios.

To further strengthen the evaluation process, human assessment may also be incorporated through manual annotation or scoring rubrics. Human evaluators can assess factual consistency, interpretability, and overall response reliability using predefined scoring criteria. This qualitative evaluation complements the quantitative metrics and provides additional insight into subtle hallucination behaviours that may not be fully captured through autonomous evaluation alone.

By combining retrieval, generation, hallucination, and efficiency metrics, the evaluation framework provides a comprehensive and complete assessment of each experimental configuration. This multi-dimensional evaluation approach enables for the accurate comparison between the baseline RAG model design and the proposed hallucination mitigation strategy while ensuring that improvements in reducing hallucinations do not negatively impact the overall system practicality and reliability.

4.0 Sustainability Considerations

This project employs a variety of sustainability principles through environmental responsibility, economic feasibility, and social responsibility throughout the design and implementation of the project. As AI systems continue to grow in complexity and become more computationally demanding, it becomes increasing crucial to design solutions that are efficient, accessible, and ethically responsible. The aim of the project is to not only create a functioning system, but to ensure that the solution remains to be resource-efficient, accessible, maintainable, and beneficial to all stakeholders involved over time.

4.1 Environmental Sustainability

Environmental sustainability is addressed through the reuse of existing AI technologies and efficient computational practices. Instead of training and developing a LLM from scratch, this project leverages pre-trained open-source models through Ollama, significantly reduce the demand of computational resources, electrical energy consumption, and hardware requirements. Ollama is an open-source tool that allows users to download, manage and run LLMs directly on the user's local hardware. Training modern AI models independently requires significant GPU processing power, electricity, cooling systems, and large-scale data centres. All these requirements contribute greatly to carbon emissions and electronic wastage. By reusing pre-trained models. The project avoids unnecessary duplication of computational efforts which leads to the significant reduction of its environmental footprint and impact on the environment.

The implementation of a Retrieval-Augmented Generation (RAG) architecture further improves the sustainability factor by allowing for the design of smaller and more efficient models that perform specialised tasks using external datasets rather than relying entirely on increasingly larger AI models. Instead of generating a completely new response solely from the model's memory, RAG systems retrieve only information relevant to the query from a vector database before generating a response. This greatly reduces overhead, improves the efficiency of the model, and vastly lowers the hardware requirements needed to run the system effectively.

This project encourages the local deployment of the model where possible, reducing continuous dependence on cloud-based inference servers. As such, local execution decreases repeated network communication and external server utilisation, contributing to requiring less electricity consumption during the operation of the model. Efficient data-processing techniques, controlled dataset sizes, and selective retrieval methods are all also used in conjunction to minimise any unnecessary computational and storage usage.

4.2 Economic Feasibility

Economic sustainability is a major consideration within the project, especially in the context of university level research applications and small-scale models. The system is designed using the widely available open-source technologies such as Python, ChromaDB, and Ollama, which eliminates licensing fees and reduces the overall costs for developing and implementing the model. This approach allows for the project financially accessible to students, researchers, and small organisations that may not have access to enterprise-level AI infrastructure.

Using pre-trained models also removes the extremely high costs usually associated with training AI systems, often requiring expensive GPU usage and long training durations. Instead, most resources can then be used to improve the system's functionality, quality of the dataset, retrieval performance, and overall user experience. Hence, local model hosting can additionally lead to the reduction of ongoing operational expenses by limiting the costs for API usages and subscription costs commonly associated with many commercial AI services.

On a long-term perspective, modularly designing the project aims to improve economic sustainability by simplifying the maintaining and updating of the model. Since the system is made of many individual components, such as embedding models, vector databases, or language models, they can all be updated independently without having to rebuild the entire system. Hence, it greatly reduces any redevelopment costs for any future work and improvements and extends the usable lifespan of the project.

4.3 Technical Sustainability

Technical sustainability is the focus on ensuring that this project will remain maintainable, scalable, and adaptable to any future advancements and developments. The system architecture is intentionally modular, separating all responsibilities, such as data-processing, embedding generation, retrieval, and response generation into multiple independent components that work cohesively together to form a fully functional system. This design improves maintainability and enables future improvements and changes to be implemented more efficiently and effectively.

For example, if the embedding model has a replacement that is newer and/or more accurate, it can be replaced without fundamentally altering the rest of the application. Similarly, the vector database or language model can be upgraded as technology improves over time. This flexible structure reduces any risk of technical components becoming obsolete and supports usability of the system in the long run beyond the scope of the initial project.

The project also prioritises the interoperability of the model through the use of standard libraries and open-source technologies. As such, it increases compatibility with any future tools and encourages reproducibility, collaboration, and future developments with other students, researchers, or developers. Modular and open-source devices contribute to the long-term sustainability and maintainability of the system.

4.4 Social Responsibility & Ethical Considerations

Social Responsibility is addressed through accessible, privacy, transparency, and the responsible use of AI. By using open-source technologies and enabling the deployment on local hardware, the project lowers barriers to entry for students and researchers that may not have access to proprietary AI systems. This is to promote a broader participation in AI development across all fields, as well as supporting more equitable access to emerging technologies.

Privacy and data security are two very important considerations that must be taken into account as well. Especially in the contexts of governance documents, certain information may not be privy to the public. As such, local execution through Ollama allows for sensitive information to remain on the device rather than being transmitted to third-party cloud services, reducing any potential risks associated with external data storage and processing. Hence, for applications involving confidential, academic, or proprietary information, local execution reduces any risk of leaking data to any unknown entities.

The project additionally incorporates responsible AI principles through the use of RAG. Since responses are generated using retrieved sources of information from a defined and accurate dataset, the system reduces the risk of hallucinations and improves response traceability compared to standalone generative AI models. This improves both transparency and reliability in the model, which are very important considerations both in academic and professional applications.

By encouraging the sharing of sustainable knowledge through the use of open-source technologies, reusable development practices, and collaborative learning resources, the project supports wider research and development in AI systems while promoting innovation that is technically effective whilst remaining socially responsible.

4.5 Overall Sustainability Impact

This project demonstrate how sustainability principles can be integrated into AI system development and continue to be an integral consideration for all implementations of AI technologies. Through efficient resource utilisation, cost-conscious design, and ethical implementation practices, the model prioritises model reuse, modularity, accessibility and responsible deployment. This project aims to deliver a practical AI solution that remains environmentally conscious, economically viable, technically maintainable, and socially beneficial

5.0 Conclusions and Project 2 Plan

In conclusion, substantial progress has been made toward the development of a hallucination-aware QA system for governance documentation. An extensive literature review was completed to investigate the causes of hallucinations in LLMs and the effectiveness of existing mitigation strategies. Following this, a baseline RAG model was successfully implemented using IEEE governance documents as the primary knowledge corpus. Additional progress includes document pre-processing, vector embedding generation, vector database implementation, retrieval pipeline development, and the creation of a governance-related benchmark question set for evaluation.

Several challenges were identified during the development of the RAG mode, particularly regarding the accurate retrieval and interpretation of highly structured governance information. Maintaining factual consistency whilst minimising hallucinations remains difficult due to the probabilistic nature of LLM generation. Furthermore, the issue of balancing improved verification methods along with computational efficiency is expected to be an ongoing challenge throughout experimentation.

Future work will focus on integrating and evaluation additional hallucination mitigation strategies, including RISE, RLVR, XAI, and multi-modal verification techniques. These approaches will be comparatively analysed using metrics such as hallucination rate, faithfulness, retrieval quality, accuracy, and computational performance. The expected outcome of this project is the development of a more reliable and trustworthy governance-oriented QA system capable of generating factually grounded and document-supported responses with reduced hallucination occurrence. In addition, a comprehensive analysis and comparison of the various mitigation strategies will be performed and displayed in an experimental report-like format explaining the outcomes.

6.0 References

[1] S. Joshi, “Comprehensive review of AI hallucinations: Impacts and mitigation strategies for financial and business applications,” International Journal of Computer Applications Technology and Research, vol. 14, no. 6, pp. 38–50, 2025, doi: 10.7753/IJCATR1406.1003.

[2] Y. Sun, D. Sheng, Z. Zhou, and Y. Wu, “AI hallucination: Towards a comprehensive classification of distorted information in artificial intelligence-generated content,” Humanities and Social Sciences Communications, vol. 11, art. no. 1278, 2024, doi: 10.1057/s41599-024-03811-x.

[3] F. Cuconasu, G. Trappolini, F. Siciliano, S. Filice, C. Campagnano, Y. Maarek, N. Tonellotto, and F. Silvestri, “The Power of Noise: Redefining Retrieval for RAG Systems,” in Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR ’24), New York, NY, USA: Association for Computing Machinery, 2024, pp. 719–729.

[4] M. Arslan, H. Ghanem, S. Munawar, and C. Cruz, “A Survey on RAG with LLMs,” Procedia Computer Science, vol. 246, pp. 3781–3790, 2024, presented at the 28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024).

[5] X. Liu, T. Liang, Z. He, J. Xu, W. Wang, P. He, Z. Tu, H. Mi, and D. Yu, “Trust, But Verify: A Self-Verification Approach to Reinforcement Learning with Verifiable Rewards,” arXiv preprint arXiv:2307.16813, 2023.

[6] X. Wen, Z. Liu, S. Zheng, Z. Xu, S. Ye, Z. Wu, X. Liang, Y. Wang, J. Li, Z. Miao, J. Bian, and M. Yang, “Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs,” arXiv preprint, Jun. 2025, doi: 10.48550/arXiv.2506.14245.

[7] “Extractive AI,” FlowHunt Glossary, FlowHunt, 2025. [Online]. Available:https://www.flowhunt.io/glossary/extractive-ai/#. [Accessed: 15 Mar. 2026].

[8] V.-H. Phan, T.-D. Le-Duc, L.-K. Pham, A.-T. Le, Q.-H. Dinh-Nguyen, D.-Q. Vo, H.-Q. Nguyen-Son, A.-D. Tran, D. Vu, and M.-S. Dao, “Fact-checking at scale: Multimodal AI for authenticity and context verification in online media,” in Proc. 33rd ACM Int. Conf. Multimedia (MM ’25), 2025, pp. 14048–14054, doi: 10.1145/3746027.3767108.

[9] R. Dwivedi, D. Dave, H. Naik, S. Singhal, O. Rana, P. Patel, B. Qian, Z. Wen, T. Shah, G. Morgan, and R. Ranjan, “Explainable AI (XAI): Core ideas, techniques, and solutions,” ACM Computing Surveys, vol. 55, no. 9, pp. 1–33, 2023, doi: 10.1145/3561048.

[10] K. Blagec, G. Dorffner, M. Moradi, and M. Samwald, “A critical analysis of metrics used for measuring progress in artificial intelligence,” arXiv preprint, 2020.

[11] C.-S. Chan, H. Kong, and G. Liang, “A Comparative Study of Faithfulness Metrics for Model Interpretability Methods,” Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL), pp. 5029–5038, 2022. doi: 10.18653/v1/2022.acl-long.345.

[12] X. Zheng, F. Shirani, Z. Chen, C. Lin, W. Cheng, W. Guo, and D. Luo, “F-Fidelity: A Robust Framework for Faithfulness Evaluation of Explainable AI,” International Conference on Learning Representations (ICLR), 2025. doi: 10.48550/arXiv.2410.02970.

[13] L. Siagian, “The Importance of Test-Case Design in AI Evaluation: A Study of RAG Recommendation Systems with Top-K Ranking Metrics,” ResearchGate, 2025. [Online]. Available: ResearchGate PDF. [Accessed: 28-May-2026].